

Design Calculations

3S6P Lithium-Ion Battery Pack — Electrical Design Basis

Project: 3S6P Li-Ion Battery Pack Design

Author: Amey Chougule

This document records the analytical calculations used to establish the electrical ratings and simplified performance model for an 18-cell 3S6P lithium-ion battery pack.

1. Design Inputs

Parameter	Value
Cell chemistry/type	18650 Lithium-Ion
Cell nominal voltage	3.7 V
Cell maximum charging voltage	4.2 V
Cell capacity	2.0 Ah
Series count (S)	3
Parallel count (P)	6
Total cells	18
Modeled pack resistance	0.020 Ω (20 mΩ)
Maximum analyzed load current	30 A

2. Series/Parallel Configuration

The selected architecture is 3S6P: three series groups, with six cells connected in parallel within each group.

Total cells = $S \times P = 3 \times 6 = 18$ cells

3. Nominal and Maximum Pack Voltage

3.1 Nominal Voltage

Series-connected cells add their voltages.

$$\begin{aligned} V_{\text{pack,nom}} &= S \times V_{\text{cell,nom}} \\ &= 3 \times 3.7 \\ &= \mathbf{11.1\text{ V}} \end{aligned}$$

3.2 Maximum Pack Voltage

For a 4.2 V maximum cell voltage:

$$V_{\text{pack,max}} = 3 \times 4.2 = \mathbf{12.6\text{ V}}$$

4. Pack Capacity

Parallel cells add their capacity while maintaining approximately the same voltage.

$$\begin{aligned}C_{\text{pack}} &= P \times C_{\text{cell}} \\&= 6 \times 2.0 \text{ Ah} \\&= \mathbf{12.0 \text{ Ah}}\end{aligned}$$

5. Nominal Energy

Nominal energy is calculated from nominal pack voltage and rated pack capacity.

$$\begin{aligned}E_{\text{nom}} &= V_{\text{pack,nom}} \times C_{\text{pack}} \\&= 11.1 \times 12.0 \\&= \mathbf{133.2 \text{ Wh}}\end{aligned}$$

The 133.2 Wh figure is a nominal energy rating. Actual usable energy depends on discharge cutoff, cell characteristics, temperature, aging, BMS behavior, and load profile.

6. Parallel-Group Resistance

For six identical cells in parallel, the idealized equivalent resistance is:

$$R_{\text{group}} = R_{\text{cell}} / P$$

The pack model used for this project is specified directly as $R_{\text{pack}} = 0.020 \, \Omega$. This value represents the simplified design model used for voltage-drop and I^2R -loss calculations. It should not be interpreted as a precision measurement of the completed pack.

Important: If individual-cell DC resistance is available, the group and pack resistance can be recalculated from measured cell resistance plus interconnect, fuse, BMS, connector, and wiring contributions.

7. Voltage Drop Under Load

The simplified resistive voltage sag is calculated using Ohm's law:

$$V_{\text{drop}} = I \times R_{\text{pack}}$$

At 30 A:

$$\begin{aligned} V_{\text{drop}} &= 30 \times 0.020 \\ &= \mathbf{0.60 \, V} \end{aligned}$$

Therefore, using the simplified model, the loaded pack voltage relative to the 11.1 V nominal reference is approximately 10.5 V at 30 A before considering additional voltage effects.

8. Resistive Power Loss

The power dissipated by the modeled pack resistance is:

$$P_{\text{loss}} = I^2 \times R_{\text{pack}}$$

At 30 A:

$$\begin{aligned} P_{\text{loss}} &= 30^2 \times 0.020 \\ &= 900 \times 0.020 \\ &= \mathbf{18 \, W} \end{aligned}$$

This loss is an electrical estimate and is useful for preliminary thermal-awareness analysis. Real heat generation may be higher because BMS MOSFETs, connections, wires, contacts, and individual cells also contribute losses.

9. Idealized Output Power

Using the nominal voltage as a reference:

$$\begin{aligned} P_{\text{nominal}} &= V_{\text{nom}} \times I \\ &= 11.1 \times 30 \\ &= \mathbf{333 \, W} \end{aligned}$$

Using the simplified loaded voltage after the modeled resistive drop:

$$\begin{aligned} V_{\text{loaded}} &= 11.1 - 0.60 = \mathbf{10.5 \, V} \\ P_{\text{load,model}} &= 10.5 \times 30 = \mathbf{315 \, W} \end{aligned}$$

10. Efficiency Calculation

For a simplified resistive model, efficiency can be estimated from useful load power relative to the nominal electrical input power:

$$\eta = P_{\text{load}} / P_{\text{nominal}} \times 100$$
$$\eta = 315 / 333 \times 100 \approx \mathbf{94.6\%}$$

This agrees with the approximately 94–95% efficiency shown in the simulation. The value is model-based and should not be treated as measured pack efficiency.

11. Current Sweep Reference Calculations

Current (A)	Voltage Drop (V)	I ² R Loss (W)	Modeled Loaded Voltage (V)
0.0	0.000	0.00	11.100
1.5	0.030	0.04	11.070
3.0	0.060	0.18	11.040
4.5	0.090	0.41	11.010
6.0	0.120	0.72	10.980
7.5	0.150	1.12	10.950
9.0	0.180	1.62	10.920
10.5	0.210	2.21	10.890
12.0	0.240	2.88	10.860
13.5	0.270	3.65	10.830
15.0	0.300	4.50	10.800
16.5	0.330	5.45	10.770
18.0	0.360	6.48	10.740
19.5	0.390	7.61	10.710
21.0	0.420	8.82	10.680
22.5	0.450	10.12	10.650
24.0	0.480	11.52	10.620
25.5	0.510	13.01	10.590
27.0	0.540	14.58	10.560
28.5	0.570	16.25	10.530
30.0	0.600	18.00	10.500

12. Design Summary

Calculated parameter	Result
Pack configuration	3S6P
Total cells	18
Nominal voltage	11.1 V
Maximum voltage	12.6 V

Calculated parameter	Result
Pack capacity	12 Ah
Nominal energy	133.2 Wh
Modeled resistance	20 mΩ
Voltage drop at 30 A	0.60 V
I ² R loss at 30 A	18 W
Modeled load voltage at 30 A	10.5 V
Modeled efficiency at 30 A	≈94.6%

13. Engineering Assumptions & Limitations

- Cells are assumed to be matched and have similar electrical characteristics.
- The 20 mΩ pack resistance is a simplified design-model value.
- Temperature dependence of cell resistance is not included in the basic equations.
- BMS MOSFET, connector, wire, and interconnect losses are not separately modeled.
- Actual capacity and energy depend on discharge current, cutoff voltage, temperature, and cell aging.
- Simulation values must be separated from experimentally measured values in project documentation.

14. Recommended Experimental Validation

For a complete engineering validation, measure pack voltage and temperature at controlled load currents, record the individual series-group voltages, and calculate actual voltage sag, usable energy, and efficiency. A calibrated electronic load and appropriate thermal monitoring are recommended for high-current characterization.